

Erin Weiss

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Portfolio: erin-weiss.github.io | github.com/Erin-Weiss

SUMMARY

Data Scientist and ML Engineer who builds tools people can run and trust. Trained gradient-boosted and deep learning models on datasets of 200k+ records and packaged them as production-ready APIs with containerized infrastructure, monitoring, and CI/CD. Delivers client-facing data applications end to end, including a live geospatial tool that searches 127k parcel records and ships an auditable methodology log with every run. Background in strategic planning and business analytics provides a unique ability to connect model outputs to measurable business outcomes. Known for exceptional organization, analytical problem-solving, and communicating complex technical concepts to diverse audiences.

SKILLS

ML & Modeling: Machine Learning, Deep Learning, MLOps, CatBoost, scikit-learn, TensorFlow, Keras, NLP, Computer Vision, CNNs, Reinforcement Learning, Recommender Systems, Time Series Analysis, Predictive Modeling, Feature Engineering, PCA, Optimization, Gurobi

MLOps & Deployment: FastAPI, Docker, Kubernetes, GitHub Actions (CI/CD), Prometheus, Grafana, Pydantic, Uvicorn, pytest, ruff, Streamlit Cloud

Data & Analysis: Python, R, SQL, pandas, NumPy, GeoPandas, Shapely, PySpark, Apache Spark, Matplotlib, Seaborn, Folium, Jupyter, Quarto, Statistical Modeling, Geospatial Analysis, Data Visualization

Databases & Data Engineering: PostgreSQL, MySQL, Microsoft SQL Server, SQLite, dbt, Hadoop, Google Cloud Dataproc, Data Warehousing, Dimensional Modeling, ETL, GeoParquet, Alteryx, SSRS

Tools & Platforms: Git, GitHub, Bash, Streamlit, Tableau, Google Analytics, LaTeX, Figma, HTML, CSS, Microsoft Office, Excel

EXPERIENCE

Innovative Data Solutions, Remote

Aug 2024 – Present

Independent Data Scientist

- Delivered a live geospatial notification tool for a recurring disaster recovery client, replacing manual parcel-by-parcel lookup with a repeatable run that returns door hanger lists, per-site walking checklists, and an interactive map for a 30-site project in about a minute
- Built and deployed a Streamlit web application for the same client to clean, standardize, and reconcile Excel invoice data, reducing manual processing time
- Developed a Python pipeline to automate recurring data exports into formatted Excel reports with conditional formatting, threshold analysis, and embedded visualizations

College of William & Mary, Remote

July 2023 – Aug 2024

Python Tutor

- Created tutorials for graduate students in the OMSBA Python course; simplified complex subjects into digestible material
- Collaborated with the professor to integrate student feedback and improve future course content
- Recruited to tutor immediately after completing the course due to outstanding performance; received consistent positive feedback from students

Spirits & Spice, Remote

Feb 2020 – Nov 2022

Director of Strategic Planning

- Gathered and analyzed data to forecast and plan key strategies for the company including shipping costs, cost of goods, sales, marketing, wages, and rent
- Designed and implemented strategies to drive traffic and sales, leading to 2x year-over-year growth on the website in the first year
- Managed marketing strategies end-to-end, allocating budgets, approving initiatives, and keeping the team on timeline
- Negotiated contracts for marketing, retail leasing, and contractors

- Created and managed timelines, deliverables, milestones, and communications strategy within the GBPC team
- Collaborated with the firm's web team to update the GBPC corporate website, translating thought-leadership material into a web-relevant format

PROJECTS

Used Car Price Prediction — Model Development | [GitHub Repo](#)

- Trained a CatBoost model predicting used-car prices within ~\$1,300 at the median across a 243k-record dataset spanning 29 manufacturers and 5,600+ model variants
- Compared three architectures (Ridge regression, CatBoost, FT-Transformer) and selected CatBoost for production based on accuracy, inference speed, and native categorical feature handling
- Engineered 20+ features from raw vehicle attributes; managed reproducible experiments with versioned run artifacts

Used Car Price Prediction API — Production Engineering | [GitHub Repo](#)

- Built the trained model into a production-ready API using FastAPI, Docker, and Kubernetes with HPA autoscaling, Prometheus + Grafana monitoring, and CI/CD via GitHub Actions
- Implemented request validation with Pydantic v2, fuzzy input matching for free-text fields, and median imputation for optional features
- Built a full test suite including unit tests, integration tests, and end-to-end predictions against the real model, running automatically on every merge to main

Demolition Neighbor Notification Tool — Geospatial Engineering | [Live App](#) | [GitHub Repo](#)

- Searched 126,958 St. Louis parcels against 500 ft buffers measured from each demolition site's boundary rather than its center, using a projected CRS and spatial indexing to return 1,636 notification addresses across a 30-site project
- Flagged 134 parcels whose assessor records contradicted each other for in-person verification, and shipped an assumptions log and match report with every run so the client could audit how each address was derived
- Generalized a single-use client script into a dual-interface application (command line and Streamlit) sharing one pipeline, matching client site lists to city records by parcel ID with an address fallback; validated by 44 tests and ruff linting on every push

EDUCATION

William & Mary, Raymond A. Mason School of Business, Williamsburg, Virginia
Master of Science in Business Analytics

Aug 2024

Georgetown University, School of Foreign Service, Washington, D.C.
Bachelor of Science, Science, Technology, and International Affairs (STIA); Minor — Chinese

May 2018

ADDITIONAL INFORMATION

Leadership: Managed teams, projects, and individuals in roles as Director of Strategic Planning and Communications Coordinator

Languages: Fluent in English and proficient in Mandarin. Passed Georgetown SFS Fluency Exam (Spring 2018); Attained HSK Certification Level II (Fall 2014) after self-teaching Chinese

Volunteer: After School Kids (A.S.K.) Volunteer, Smithsonian Volunteer